



DEPARTMENT OF ELECTRONICS AND COMPUTER ENGINEERING

Experiment No.: 3

Interfacing of LPC2148 with LEDs.

Name: _____

Roll No.: _____ *Batch:* _____

Date of Performance: _____

Date of Assessment: _____

Particulars	Marks
Marks for Regularity (05)	
Marks for Performance (15)	
Understanding (Viva) (05)	
Total (25)	
Signature of Course Teacher	

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Experiment: 3

AIM: Interfacing of LPC2148 with LEDs.

OBJECTIVE:

1. To understand the basic concept of LED interfacing with a microcontroller.
2. To learn how to control an LED using GPIO pins of LPC2148.
3. To study the behavior of digital output (HIGH/LOW) signals.

APPARATUS:

1. LPC2148 module development board.
2. Connecting wire.
3. Adapter.

THEORY:

Light Emitting Diode (LED) is a semiconductor device that emits light when current flows through it in the forward direction. It is a p-n junction diode that converts electrical energy into light energy (electroluminescence). LEDs are widely used as indicators, displays, and in communication systems due to their low power consumption, long life, and small size.

Applications of LED:

1. Picture phones and digital watches
2. Camera flashes and automotive heat lamps
3. Aviation lighting
4. Digital computers and calculators
5. Traffic signals and Burglar alarms systems
6. Microprocessors and multiplexers
7. Optical Communication
8. Indicator lamps in electric equipment
9. LED television
10. Vehicle head lamps, domestic and decorative illumination, street lighting.

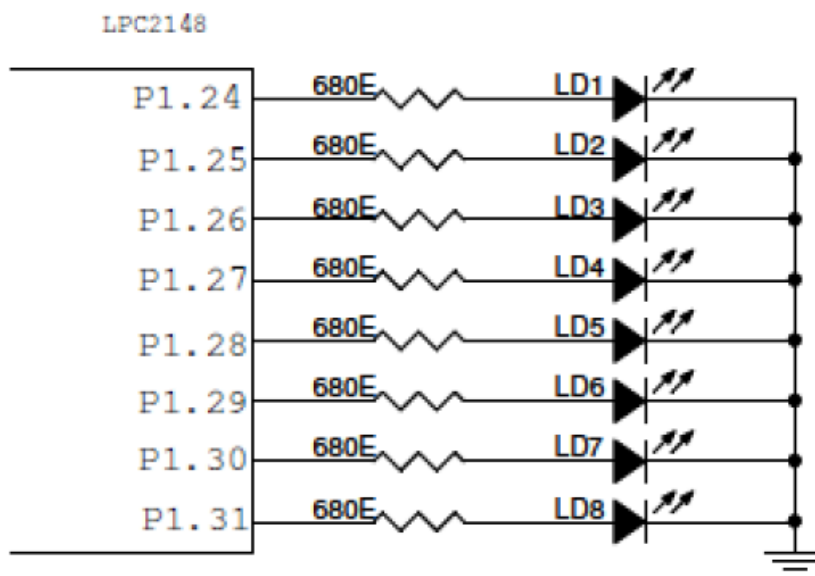
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GPIO (General Purpose Input/Output) in LPC2148:

The LPC2148 has **two GPIO ports: Port 0 (P0) and Port 1 (P1)**, each with up to 32 pins. GPIO pins can be configured as digital inputs or outputs. For LED interfacing, the pins are configured as **outputs**. The direction and data are controlled through memory-mapped registers.

Generic Name	Description	Access	Reset value ^[1]	PORT0 Address & Name	PORT1 Address & Name
IOPIN	GPIO Port Pin value register. The current state of the GPIO configured port pins can always be read from this register, regardless of pin direction.	R/W	NA	0xE002 8000 IO0PIN	0xE002 8010 IO1PIN
IOSET	GPIO Port Output Set register. This register controls the state of output pins in conjunction with the IOCLR register. Writing ones produces highs at the corresponding port pins. Writing zeroes has no effect.	R/W	0x0000 0000	0xE002 8004 IO0SET	0xE002 8014 IO1SET
IODIR	GPIO Port Direction control register. This register individually controls the direction of each port pin.	R/W	0x0000 0000	0xE002 8008 IO0DIR	0xE002 8018 IO1DIR
IOCLR	GPIO Port Output Clear register. This register controls the state of output pins. Writing ones produces lows at the corresponding port pins and clears the corresponding bits in the IOSET register. Writing zeroes has no effect.	WO	0x0000 0000	0xE002 800C IO0CLR	0xE002 801C IO1CLR

INTERFACING DIAGRAM:



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PIN DESCRIPTION:

LED	Pin Details	
	LPC2148	Description
LD 0	P1.24	Common Cathode LED 0
LD 1	P1.25	Common Cathode LED 1
LD 2	P1.26	Common Cathode LED 2
LD 3	P1.27	Common Cathode LED 3
LD 4	P1.28	Common Cathode LED 4
LD 5	P1.29	Common Cathode LED 5
LD 6	P1.30	Common Cathode LED 6
LD 7	P1.31	Common Cathode LED 7

CONCLUSION:
